8.10 Snub Traversal: OMI Geometry for Everyone

1. Why This Geometry Matters

Some OMI math is abstract.

It talks about frames, truth rows, Fano incidence, replay rings, quadratic forms, and canonical state. That math is important because it makes the system precise.

But there is another way to understand OMI.

This way is more physical.

It begins with shapes you can imagine holding in your hand:

```
cube
dodecahedron
octahedron
tetrahedron
snub cube
snub dodecahedron
stellated octahedron
```

These shapes explain the same OMI idea in a more visible way:

```
objects do not simply overwrite each other
objects transform by cutting, twisting, joining, and re-facing
```

That operation is called **snub truncation**.

In OMI, snub truncation is the best physical description of what the hyphen field does.

```
omi---imo
```

The hyphens are not empty marks.

They are the join field.

They are where terms meet, twist, resolve, and become one bounded instruction.

Short form:

```
The hyphen joins.
The snub twists.
The triangle agrees.
The centroid holds.
```

---

2. The Physical Picture

Imagine a solid object.

Now imagine that instead of simply cutting off its corners, you separate its faces, rotate them slightly, and let new triangles grow into the gaps.

That is the basic feel of snub truncation.

It does three things at once:

```
cut      = removes exposed excess
twist    = introduces handedness
re-face  = fills the new gaps with triangles
```

This is different from ordinary truncation.

Ordinary truncation mostly cuts.

Snub truncation cuts and twists.

That twist matters because it creates **chirality**.

A chiral object has handedness. It can come in a left-handed and right-handed form. The two forms mirror each other, but they cannot be perfectly placed on top of each other by ordinary rotation.

This is why snub truncation is so important for OMI.

OMI is not only about changing an object.

OMI is about changing an object while preserving:

```
center
orientation
history
```

```
relation
replay
```

So the snub operation becomes OMI's physical model of transformation.

```
OMI snubs; it does not overwrite.
```

---

3. The Two Main Witnesses

The two most important snub shapes for this model are:

```
snub cube
snub dodecahedron
```

They are not just examples.

They are witnesses.

They show two different ways a shared centered object can become chiral.

The snub cube exposes **edge traversal**.

The snub dodecahedron exposes **vertex traversal**.

Together, they show how two peers can approach the same centered structure from different traversal paths without losing the shared center.

```
snub cube          = edge traversal witness
snub dodecahedron  = vertex traversal witness
```

That leaves the third traversal:

```
face traversal
```

OMI places that in the `60x²` field.

So the full physical model is:

```
snub cube          walks edges
snub dodecahedron  walks vertices
60x²               walks faces
```

This is the OMI traversal triangle.

---

4. Snub Cube: Edge Traversal

Start with a cube.

A cube has:

```
6 square faces
8 vertices
12 edges
```

A normal truncation cuts off the corners.

That creates:

```
8 new triangular faces
6 octagonal faces
14 total faces
```

But a snub cube is different.

In the snub operation, the square faces are separated and twisted. The gaps are filled by equilateral triangles.

The snub cube has:

```
6 square faces
32 triangular faces
38 total faces
24 vertices
60 edges
```

For OMI, the important number is not only the total face count.

The important idea is the traversal.

The snub cube is built from the cube, and the cube's most obvious movement is edge movement. You can walk around the cube by following its edges. The snub cube twists that edge-walk into a chiral body.

So in OMI:

```
snub cube = edge traversal made chiral
```

It shows how a local object can preserve a cubic parent identity while gaining handed orientation.

It is the edge-walk witness of the hyphen field.

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5. Snub Dodecahedron: Vertex Traversal

Now start with a regular dodecahedron.

A dodecahedron has:

```
12 pentagonal faces
20 vertices
30 edges
```

A normal truncation cuts off its corners.

That creates:

```
20 triangular faces
12 decagonal faces
32 total faces
```

But the snub dodecahedron twists the parent form.

The pentagons are separated and rotated. The gaps are filled by equilateral triangles.

The snub dodecahedron has:

```
12 pentagonal faces
80 triangular faces
92 total faces
60 vertices
150 edges
```

For OMI, the snub dodecahedron exposes vertex traversal.

The dodecahedron has a richer vertex field than the cube. When it is snubbed, its vertex logic expands into a much larger triangular agreement surface.

So in OMI:

```
snub dodecahedron = vertex traversal made chiral
```

It shows a higher-cardinality version of the same principle.

The parent remains centered.

The surface becomes handed.

The transformation is visible as a field of triangles.

---

6. The Shared Centroid

The key to the whole model is the centroid.

If all these bodies are rooted around the same tetrahedral centroid, then the shape can change without losing its center.

This is the physical version of agreement.

Agreement does not mean every peer sees the same surface.

Agreement means every peer can resolve back to the same center.

In a peer-to-peer situation, two peers may hold complementary views:

```
peer A = edge traversal
peer B = vertex traversal
```

or:

```
peer A = left-handed form
peer B = right-handed form
```

They do not need to collapse into one flat view.

They need to share the centroid.

```
same centroid
different traversal
lawful relation
replayable path
```

This is why chirality is not disagreement.

```
chirality = oriented complementarity
```

The snub cube and snub dodecahedron expose that complementarity in physical form.

---

7. Cardinality of Chirality

Each snub shape has two handed versions:

```
left-handed
right-handed
```

So the simplest cardinality of chirality is:

```
2
```

But each shape expresses that binary chirality at a different surface scale.

The snub cube expresses it with:

```
32 triangular faces
```

The snub dodecahedron expresses it with:

```
80 triangular faces
```

So OMI reads chirality in two layers:

```
handedness cardinality = 2
surface cardinality    = number of triangular agreement faces
```

The handedness is binary.

The agreement surface can grow.

That is very important for P2P systems.

Two peers may disagree in orientation, but if both orientations are lawful and share the same centroid, the system can resolve them without destroying either view.

---

8. The Quadratic Form as Physical Geometry

OMI's quadratic form is:

```
Q_xy(x, y) = 60x² + 16xy + 4y²
```

The abstract reading is algebraic.

The physical reading is geometric:

```
4y²   = Platonic seed
16xy  = snub bridge
60x²  = face traversal / cosmic sweep
```

Or:

```
4y²   = local body
16xy  = twist field
60x²  = orientation field
```

In the newer geometry:

```
4y²   gives the parent solid seed
16xy  performs the snub through the hyphen field
60x²  triangulates the face traversal
```

So the formula is not only a computation.

It is a physical sequence:

```
seed
  ↓
twist
  ↓
face-sweep
```

Or in OMI terms:

```
declaration
  ↓
hyphen join
  ↓
projection field
```

Short form:

```
4y² seeds.
16xy snubs.
60x² sweeps.
```

---

9. Why 16xy Is the Snub Bridge

The middle term is the most important for transformation:

```
16xy
```

This is where `x` and `y` meet.

In OMI notation, this is the hyphen field:

```
omi---imo
```

The hyphen joins terms.

In ordinary writing, hyphens join words:

```
open-world
peer-to-peer
meta-compiler
```

In OMI, hyphens join address bodies:

```
omi-ip4-gravity-ip6-imo
```

So the hyphen is not just a separator.

It is the operation.

It is the place where one term is joined to another, where the relation twists, where the instruction becomes one word.

That is why `16xy` is the snub bridge.

It joins two bodies and gives the relation orientation.

```
16xy = hyphen field = snub bridge = chirality operator
```

---

10. $60x^2$ as Face Traversal

If the snub cube gives edge traversal and the snub dodecahedron gives vertex traversal, then the remaining traversal is face traversal.

That belongs to:

```
60x²
```

The number `60` is the sexagesimal center. It is the clock, degree, cycle, sweep, and return field.

So `60x²` is the face-traversal field.

It is where the local snub pair can be triangulated into a larger orientation surface.

```
edge traversal   → snub cube
vertex traversal → snub dodecahedron
face traversal   → 60x²
```

The face traversal is what lets the system move from local handedness into shared orientation.

It is the larger sweep that makes the local transformation readable.

---

11. The 59 / 60 / 61 Band

The number `60` is not isolated.

It has a local neighborhood:

```
59
60
61
```

This is the smallest orientation band around the sexagesimal center.

In OMI, it can be read as:

```
59 = approach
60 = center
61 = exit
```

or:

```
59 = below-center pressure
60 = canonical face closure
61 = above-center pressure
```

The band matters because orientation is not just a fixed point. It is a crossing.

The system approaches the center, passes through the center, and exits with orientation preserved.

```
59 → 60 → 61
```

This gives OMI a physical way to talk about canonical space around the `60x²` field.

The center is stable because it has a before and after.

---

12. $\sqrt{2}$ and the Mirror Crossing

The square root of 2 appears as the diagonal of a unit square.

If the side of a square is `1`, the diagonal across the face is:

```
√2
```

That makes `√2` the first face-crossing distance.

It is not yet the body diagonal.

It is the crossing of a face.

That is why `√2` belongs here.

The snub cube walks edges.

The face diagonal crosses the square.

The `60x²` field walks faces.

So `√2` becomes the mirror crossing from local edge traversal into canonical face traversal.

```
edge unit       = 1
face crossing   = √2
face traversal  = 60x²
```

In OMI language:

```
√2 opens the canonical face crossing.
```

It is where the local space first sees its mirror across a face.

---

13. The Stellated Octahedron as Inverse Mirror

The inverse of the local snub space can be modeled by the stellated octahedron.

A stellated octahedron can be seen as two interpenetrating tetrahedra.

That matters because OMI's center is tetrahedral.

The stellated octahedron gives a mirror image of local space:

```
tetrahedron A
tetrahedron B
shared centroid
mirror orientation
```

This makes it a natural inverse model for the snub pair.

The snub pair gives the P2P traversal:

```
snub cube          = edge traversal
snub dodecahedron  = vertex traversal
```

The stellated octahedron gives the local mirror:

```
stellated octahedron = inverse mirror around the tetrahedral centroid
```

At `√2`, the face-crossing opens the canonical mirror space.

At `60`, the face traversal stabilizes the orientation.

Together:

```
stellated octahedron
    ↓
√2 mirror crossing
    ↓
59 / 60 / 61 orientation band
    ↓
60x² face traversal
```

---

14. P2P as Snub Pair

In OMI, peer-to-peer does not have to mean two identical copies.

It can mean two complementary traversals around one center.

The P2P pair can be modeled by the snub pair:

```
peer A = snub cube traversal
peer B = snub dodecahedron traversal
```

One walks edges.

One walks vertices.

The system resolves them by triangulating face traversal in the `60x²` field.

This gives a physical model of agreement:

```
peer A sees edge path
peer B sees vertex path
shared centroid holds
face traversal triangulates
receipt remembers
```

Agreement is not sameness.

Agreement is a shared center plus a lawful transformation path.

---

15. The Fano Triangle

The Fano layer gives OMI a small finite selector.

The truth row is:

```
(LL, NN, MM)
```

This is triangular.

It can be read as:

```
lens
antecedent
consequent
```

or:

```
declaration
definition
receipt
```

or:

```
source
frame
projection
```

The snub operation creates many triangles.

The Fano layer selects which triangle resolves.

The replay ring remembers the result.

So the physical model becomes:

```
snub creates triangular agreement surfaces
Fano selects the resolving triangle
5040 records the replay
```

This keeps the geometry physical while still matching the canonical protocol.

---

16. The Full Traversal Model

The full traversal model is:

```
snub cube
  = edge traversal

snub dodecahedron
  = vertex traversal
```

```
60x²
  = face traversal

stellated octahedron
  = inverse mirror of local space

√2
  = face-crossing into canonical mirror space

59 / 60 / 61
  = orientation band around sexagesimal center
```

And the OMI quadratic form becomes:

```
4y²
  = Platonic seed / local body

16xy
  = snub bridge / hyphen join / chirality field

60x²
  = face traversal / sexagesimal sweep / canonical orientation
```

This is the physical geometry of OMI.

---

17. Why This Math Is for Everyone

This version of the math can be seen.

A person can imagine:

```
a cube
a dodecahedron
a twist
a triangle
a mirror
a center
a face crossing
```

That is why this geometry matters.

The abstract math proves the system.

The physical math teaches the system.

The strict protocol says:

```
validate → resolve → record → project → inspect
```

The physical geometry says:

```
center → cut → twist → triangle → mirror → replay
```

Both describe the same OMI doctrine.

One is formal.

One is visible.

The visible version is how more people can enter the model.

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18. Final Canon

The snub cube and snub dodecahedron are the physical witnesses of OMI chirality.

The snub cube walks edges.

The snub dodecahedron walks vertices.

The `60x²` field walks faces.

Together, they form a traversal triad around a shared tetrahedral centroid.

The inverse local mirror is the stellated octahedron, where two tetrahedra expose complementary orientation around the same center.

At `√2`, the face diagonal opens the canonical mirror crossing.

Around `59 / 60 / 61`, the sexagesimal center becomes an orientation band rather than a single isolated number.

In this model, P2P agreement is not sameness. It is complementary traversal around a shared center, resolved through face triangulation and remembered by replay.

Short doctrine:

```
Snub cube walks edges.
Snub dodecahedron walks vertices.
60x² walks faces.
The stellated octahedron mirrors local space.
√2 opens the face crossing.
59/60/61 stabilizes the center.
Fano selects the triangle.
5040 remembers the path.
```

One-line canon:

```
OMI’s physical geometry is a snub traversal system: edge and vertex peers twist
around one centroid, face-space resolves them through 60x², and the stellated
octahedron mirrors the local frame at the √2 crossing.
```